Synthetic data as vicinal gradient coverage: a gauge-invariant η/L gain from rough-mode suppression

Mechanism theory for the diffusion-data robustness finding (author-derived; verified)

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Abstract

The observation is that adding diffusion-generated data to adversarial training increases the *scale-invariant* robustness ratio $\eta/L = M/L$ (margin over local sensitivity $L = \mathbb{E}\|\nabla_x M\|$), and hence the certified radius, even though the *raw* logit margin *decreases*. This note gives a static mechanism. First-order adversarial training is input-gradient regularization (Lemma 1); synthetic data changes the measure on which the penalty is enforced, adding a positive-semidefinite *vicinal gradient-coverage matrix*; increasing that matrix suppresses high-gradient modes (Theorem 1); and if a high-gradient mode also carries margin, the scale-invariant ratio η/L strictly rises exactly when a smoother low-gradient component remains (Theorem 2). The mechanism's *gauge-free* predictions are confirmed at multi-seed: η/L rises monotonically and $L_1/L_2 \approx \mathrm{const}$ while both fall (Corollary 2; $25.5 \to 23.6$). **One caveat, made precise here (Remark 2):** the finer statement "margin falls, sensitivity falls faster" is a decomposition in the model's natural weight-norm gauge ($A = I$); it is *not* gauge-invariant at the network level. Empirically it survives a logit-norm anchor (the analogue of that gauge) but flips sign under an NLL/temperature anchor, so only η/L and the radius are reported as gauge-free conclusions. The slogan, stated invariantly: *diffusion data suppresses high-gradient margin-carrying modes, raising the scale-invariant ratio η/L.*

The robust radius admits the first-order decomposition $\Delta \log r \approx \Delta \log M - \Delta \log L$. The *difference* $\Delta \log(M/L)$ is invariant under rescaling the output logits (a temperature gauge), whereas the two terms separately are not; the gauge-free content of the mechanism is therefore the rise of M/L, and the decomposition into "smaller margin" versus "smaller L" is meaningful only once a gauge is fixed (Remark 2). The mechanism is not that synthetic data raises the margin: it is that adversarial training is, to first order, input-gradient regularization, and synthetic data changes the measure on which this regularizer acts, adding a positive-semidefinite vicinal gradient-coverage matrix. Modes that carry margin but have high input-gradient energy are suppressed, and the scale-invariant proxy M/L rises.

1 First-order adversarial training is input-gradient regularization

Lemma 1 (First-order robust-loss expansion). *Let $\ell : \mathbb{R}^d \to \mathbb{R}$ be twice continuously differentiable on the closed ball $B_\varepsilon(x) = \{x + \delta : \|\delta\| \le \varepsilon\}$ for a norm $\|\cdot\|$ with dual $\|\cdot\|_*$, and suppose $\|\nabla_x^2 \ell(z)\|_{\mathrm{op}} \le H$ for all $z \in B_\varepsilon(x)$ (operator norm induced by $\|\cdot\|$). Then*

$$\sup_{\|\delta\| \le \varepsilon} \ell(x+\delta) = \ell(x) + \varepsilon \|\nabla \ell(x)\|_* + R_\varepsilon(x), \qquad |R_\varepsilon(x)| \le \tfrac{H}{2}\varepsilon^2.$$

Proof. Set $g = \nabla \ell(x)$. For $\|\delta\| \le \varepsilon$, Taylor with remainder gives $\ell(x+\delta) = \ell(x) + \langle g, \delta\rangle + \frac{1}{2}\delta^\top \nabla^2 \ell(x+s\delta)\delta$ for some $s \in [0,1]$, and $|\frac{1}{2}\delta^\top \nabla^2 \ell(x+s\delta)\delta| \le \frac{H}{2}\|\delta\|^2 \le \frac{H}{2}\varepsilon^2$. With $\langle g, \delta\rangle \le \|g\|_* \|\delta\| \le \varepsilon \|g\|_*$, the supremum obeys $\sup_{\|\delta\|\le\varepsilon} \ell(x+\delta) \le \ell(x) + \varepsilon\|g\|_* + \frac{H}{2}\varepsilon^2$. For the lower bound pick v with $\|v\| \le 1$ and $\langle g, v\rangle = \|g\|_*$ (the unit ball is compact, so the linear functional attains its max), and set $\delta = \varepsilon v$. Taylor gives $\ell(x+\varepsilon v) \ge \ell(x) + \varepsilon\|g\|_* - \frac{H}{2}\varepsilon^2$, and the supremum is at least this. Combining the two bounds gives $|\sup_\delta \ell(x+\delta) - \ell(x) - \varepsilon\|g\|_*| \le \frac{H}{2}\varepsilon^2$. $\square$

Remark 1 (From loss-gradient to margin-gradient)**.** If $\ell(x,y) = \phi(M(x,y))$ then $\nabla_x \ell = \phi'(M)\nabla_x M$, and on any margin band where $0 < c_- \le |\phi'(M)| \le c_+ < \infty$ the loss-gradient and margin-gradient penalties are equivalent up to constants: $c_-\|\nabla_x M\| \le \|\nabla_x \ell\| \le c_+\|\nabla_x M\|$. Thus the first-order adversarial penalty controls the same local sensitivity $L = \mathbb{E}\|\nabla_x M\|$ that enters the radius proxy.

2 The vicinal gradient-coverage matrix

Linearize the margin in parameters, $M_\theta(x) = \theta^\top \psi(x)$ with a differentiable feature map $\psi : \mathcal{X} \to \mathbb{R}^p$, and write $J_\psi(x) = \nabla_x \psi(x) \in \mathbb{R}^{p\times d}$, so that $\nabla_x M_\theta(x) = J_\psi(x)^\top \theta$ and $\|\nabla_x M_\theta(x)\|_2^2 = \theta^\top J_\psi(x) J_\psi(x)^\top \theta$. For a (data or vicinal) measure ν define the gradient-coverage matrix

$$Q_\nu := \int J_\psi(x) J_\psi(x)^\top \, d\nu(x), \qquad \int \|\nabla_x M_\theta(x)\|_2^2 \, d\nu(x) = \theta^\top Q_\nu \theta.$$

Q_ν is positive semidefinite, since $v^\top Q_\nu v = \int \|J_\psi(x)^\top v\|_2^2 \, d\nu(x) \ge 0$. Adding synthetic data changes ν, adding a positive-semidefinite Q_{syn} to the input-gradient penalty.

3 Spectral rough-mode filtering

Theorem 1 (Synthetic gradient coverage filters high-sensitivity modes)**.** *Let $A \in \mathbb{R}^{p\times p}$ be symmetric positive definite, $Q \in \mathbb{R}^{p\times p}$ symmetric positive semidefinite, $a \in \mathbb{R}^p$, and for $t \ge 0$ set $B_t = A + tQ$ and $\mathcal{J}_t(\theta) = \frac{1}{2}\theta^\top B_t \theta - a^\top \theta$ with minimizer θ_t. With $\eta(t) = a^\top \theta_t$ and $S(t) = \theta_t^\top Q \theta_t$,*

$$\eta'(t) = -S(t) \le 0, \qquad S'(t) = -2\,\theta_t^\top Q B_t^{-1} Q \theta_t \le 0.$$

Proof. B_t is symmetric positive definite for $t \ge 0$, so $\mathcal{J}_t$ is strictly convex with unique minimizer $\theta_t = B_t^{-1}a$ ($B_t\theta_t = a$). Differentiating $B_t\theta_t = a$ in t and using $B_t' = Q$ gives $Q\theta_t + B_t\theta_t' = 0$, i.e. $\theta_t' = -B_t^{-1}Q\theta_t$. Then $\eta'(t) = a^\top\theta_t' = (B_t\theta_t)^\top\theta_t' = \theta_t^\top B_t\theta_t' = \theta_t^\top(-Q\theta_t) = -S(t)$, and $S(t) = \theta_t^\top Q\theta_t \ge 0$. Also $S'(t) = 2(\theta_t')^\top Q\theta_t = 2(-B_t^{-1}Q\theta_t)^\top Q\theta_t = -2\theta_t^\top Q B_t^{-1} Q\theta_t = -2z^\top B_t^{-1} z$ with $z = Q\theta_t$, and $z^\top B_t^{-1} z \ge 0$ since $B_t^{-1} \succ 0$. $\square$

4 A two-mode theorem: margin down, sensitivity down more, radius up

Theorem 2 (Smoothness-not-margin in a two-mode model)**.** *Let $\theta = (\theta_s, \theta_r) \in \mathbb{R}^2$ (smooth mode s, rough mode r), $A = I_2$, $Q = \left(\begin{smallmatrix} 0 & 0 \\ 0 & q \end{smallmatrix}\right)$ with $q > 0$, and $a = (a_s, a_r)$ with $a_r \ne 0$. For $t \ge 0$ let θ_t minimize $\frac{1}{2}\theta^\top(I_2 + tQ)\theta - a^\top\theta$, and set $\eta(t) = a^\top\theta_t$, $L(t) = \sqrt{\theta_t^\top Q\theta_t}$, $\rho(t) = \eta(t)/L(t)$. Then*

$$\theta_t = \Big(a_s,\ \tfrac{a_r}{1+tq}\Big), \quad \eta(t) = a_s^2 + \tfrac{a_r^2}{1+tq}, \quad L(t) = \tfrac{\sqrt{q}\,|a_r|}{1+tq}, \quad \rho(t) = \tfrac{(1+tq)a_s^2 + a_r^2}{\sqrt{q}\,|a_r|},$$

$$\eta'(t) = -\tfrac{qa_r^2}{(1+tq)^2} < 0, \qquad L'(t) = -\tfrac{q\sqrt{q}\,|a_r|}{(1+tq)^2} < 0, \qquad \rho'(t) = \tfrac{\sqrt{q}\,a_s^2}{|a_r|} > 0 \quad (a_s \neq 0),$$

and, whenever $a_s \neq 0$, $-\frac{d}{dt}\log L(t) > -\frac{d}{dt}\log\eta(t)$. *The raw margin decreases, the sensitivity decreases faster, and the radius proxy* η/L *strictly increases.*

Proof. $(I_2 + tQ) = \mathrm{diag}(1, 1+tq)$, so $(I_2+tQ)\theta_t = a$ gives $\theta_{s,t} = a_s$ and $\theta_{r,t} = a_r/(1+tq)$. Then $\eta(t) = a_s \cdot a_s + a_r \cdot \frac{a_r}{1+tq} = a_s^2 + \frac{a_r^2}{1+tq}$. $Q\theta_t = (0, qa_r/(1+tq))^\top$, so $\theta_t^\top Q\theta_t = \frac{a_r}{1+tq}\cdot\frac{qa_r}{1+tq} = \frac{qa_r^2}{(1+tq)^2}$ and $L(t) = \sqrt{q}|a_r|/(1+tq)$ (as $1+tq>0$). Dividing, $\rho(t) = \left(a_s^2 + \frac{a_r^2}{1+tq}\right)\cdot\frac{1+tq}{\sqrt{q}|a_r|} = \frac{(1+tq)a_s^2 + a_r^2}{\sqrt{q}|a_r|}$. Differentiating: $\eta'(t) = a_r^2 \cdot (-q)(1+tq)^{-2} = -\frac{qa_r^2}{(1+tq)^2} < 0$; $L'(t) = \sqrt{q}|a_r| \cdot (-q)(1+tq)^{-2} = -\frac{q\sqrt{q}|a_r|}{(1+tq)^2} < 0$; the numerator of ρ has t-derivative qa_s^2 over the constant denominator, so $\rho'(t) = \frac{qa_s^2}{\sqrt{q}|a_r|} = \frac{\sqrt{q}a_s^2}{|a_r|} > 0$ when $a_s \neq 0$. For the logarithmic comparison, $\log L(t) = \log(\sqrt{q}|a_r|) - \log(1+tq)$ gives $-\frac{d}{dt}\log L = \frac{q}{1+tq}$, and $-\frac{d}{dt}\log\eta = \frac{qa_r^2}{(1+tq)^2\eta(t)}$. The inequality $\frac{q}{1+tq} > \frac{qa_r^2}{(1+tq)^2\eta(t)}$ is, after multiplying by $(1+tq)^2\eta(t)/q > 0$, equivalent to $(1+tq)\eta(t) > a_r^2$, i.e. $(1+tq)a_s^2 + a_r^2 > a_r^2$, i.e. $(1+tq)a_s^2 > 0$, i.e. $a_s \neq 0$. □

Corollary 1 (Pure temperature shrinkage is the degenerate case)**.** *If* $a_s = 0$ *in Theorem 2, then* $\rho(t) = |a_r|/\sqrt{q}$ *is constant: suppressing only a single rough margin-carrying mode scales margin and sensitivity together and cannot improve the radius proxy. A genuine radius gain requires a remaining smooth component* $a_s \neq 0$.

Proof. Set $a_s = 0$: $\rho(t) = \frac{a_r^2}{\sqrt{q}|a_r|} = \frac{|a_r|}{\sqrt{q}}$, independent of t, so $\rho'(t) = 0$. □

Corollary 2 (Uniform gradient-shape shrinkage)**.** *If the gradient field is dominated by the rough mode,* $\nabla_x M_{\theta_t}(x) \approx \theta_{r,t}\, g_r(x)$ *with* g_r *independent of* t*, then for any two homogeneous norms* $\|\cdot\|_\alpha, \|\cdot\|_\beta$*,*

$$\frac{\|\nabla_x M_{\theta_t}(x)\|_\alpha}{\|\nabla_x M_{\theta_t}(x)\|_\beta} \approx \frac{\|g_r(x)\|_\alpha}{\|g_r(x)\|_\beta},$$

approximately independent of t*:* L_1 *and* L_2 *can both drop sharply while* L_1/L_2 *stays nearly constant.*

Proof. Homogeneity gives $\|\nabla_x M_{\theta_t}(x)\|_\alpha \approx |\theta_{r,t}|\,\|g_r(x)\|_\alpha$ and likewise for β; since $\theta_{r,t} = a_r/(1+tq) \neq 0$, $|\theta_{r,t}|$ cancels in the ratio. □

Remark 2 (Gauge status of the decomposition)**.** The map $\theta \mapsto c\theta$ $(c>0)$ scales $M_\theta \mapsto cM_\theta$, hence $\eta \mapsto c\eta$ and $L \mapsto cL$, so the ratio $\rho = \eta/L$ is *invariant* while η and L individually are not. A trained network has exactly this freedom (post-hoc logit/temperature rescaling), so only $\rho = \eta/L$ and the input-space certified radius are gauge-free; the separate signs $\eta'(t) < 0$ and $L'(t) < 0$ in Theorem 2 are stated in the model's natural gauge, the one pinned by $A = I$ (the weight-norm / ridge normalization). The matching empirical gauge is the *logit-norm anchor*, under which the multi-seed experiment reproduces the theorem's signs (margin and L both fall, L faster); under an *NLL/temperature anchor* the split inverts. This is not a contradiction: the gauge-invariant prediction $\rho'(t) > 0$ holds under every anchor, and the decomposition is a finer, gauge-relative reading of the same inequality $-\frac{d}{dt}\log L > -\frac{d}{dt}\log\eta$. Reported conclusions are confined to the gauge-invariant layer; the decomposition is presented only relative to a stated anchor.

5 Interpretation

The smooth mode s is a low-sensitivity component that generalizes across the vicinity of the data manifold; the rough mode r is a high-sensitivity component that can raise the raw logit margin on the finite real training set. Synthetic data increases the vicinal coverage tq on the rough mode, so $\theta_{r,t} = a_r/(1+tq)$ shrinks: the raw margin $\eta(t) = a_s^2 + a_r^2/(1+tq)$ falls (the rough mode contributed margin), but the sensitivity $L(t) = \sqrt{q}|a_r|/(1+tq)$ falls faster (it is entirely the rough mode), so the radius proxy $\rho(t)$ rises whenever $a_s \neq 0$. A pure logit-temperature rescaling would shrink margin and sensitivity by the same factor and leave M/L unchanged; that is the degenerate $a_s = 0$ case. The observed increase in M/L corresponds to $a_s \neq 0$: a smooth component remains while synthetic gradient coverage suppresses rough sensitivity.

6 Adversarial verification

Remark 3 (Scope and falsifiable predictions)**. (1) Not every synthetic data helps.** A useful radius gain needs Q to penalize a rough mode more than a smooth margin-carrying one; if $Q = qI$ all modes scale uniformly and the mechanism degenerates to temperature-like shrinkage (no forced gain). **(2) It explains margin shrink, not denies it:** $\eta'(t) = -S(t) \leq 0$. **(3) It explains the radius gain with a smaller raw margin:** the inequality $-\frac{d}{dt}\log L > -\frac{d}{dt}\log\eta$ (equivalently $\rho'(t) > 0$) holds iff $a_s \neq 0$. In a fixed gauge this reads "L falls faster than M"; only the ratio statement is anchor-free (Remark 2). **(4) It is not a robust-overfitting theorem:** the proof uses no time/epochs/learning-rate decay; it is a static regularization-path result. If the effect appears early and varies monotonically with synthetic diversity, this is the mechanism; if it appears only after a late robust-overfitting phase, a separate dynamical theorem is needed. **(5) Local, not global:** the quadratic model is a local linearization; the theorem identifies the algebraic condition for the observed decomposition, not a claim that every diffusion-trained net optimizes this exact objective. **(6) Falsifiable predictions:** L decreases monotonically with synthetic diversity; M may decrease; M/L increases when a smooth component remains; $L_1/L_2 \approx$ const when the same rough gradient shape is suppressed in amplitude (Corollary 2); and the largest L drop should occur on vicinal/interpolation points where the new synthetic coverage differs most from the empirical measure.

Empirical status (multi-seed, 4 doses $\times$3 seeds). The *gauge-free* predictions hold, monotonically and with tight seed spread: η/L rises ($0.864 \to 0.984 \to 1.021 \to 1.016$; it crosses 1 and saturates by 500k), the input-space DDN ℓ_2 radius rises ($0.771 \to 0.865$), AutoAttack rises ($0.342 \to 0.468$, with AA $\leq$ PGD at every dose, i.e. no gradient masking), and the gradient-shape signature $L_1/L_2 \approx$ const is confirmed ($25.5 \to 23.6$ while L_1, L_2 roughly halve). The *gauge-relative* decomposition behaves exactly as Remark 2 predicts: under the logit-norm anchor "M falls, L falls faster" survives ($\text{margin}_{\text{ctrl}} = -0.30$, $L_{2,\text{ctrl}} = -0.46$ at 1M), whereas under the NLL/temperature anchor it inverts; only η/L and the radius are reported as conclusions. Saturation by 500k (a gain that levels off as further synthetic data adds little new vicinal coverage) is consistent with Q_{syn} saturating in coverage, a refinement beyond the single-mode model, whose $\rho(t)$ is monotone in the coverage parameter t rather than in raw sample count. The decisive remaining test is the vicinal-concentration prediction (largest L drop on/near synthetic points), the `manifold` measurement, which is *gauge-robust*: it compares L across point-sets within one fixed network, so the output gauge cancels.